ArgOS: a resource governor for a re-configurable vessel

Team entry for Fincantieri's **Software Defined Vessel** challenge (Maritime Dual-Use Hackathon, San Francisco, 16-18 Oct 2026). ArgOS, after the Argo, is the operating system the ship does not have: it owns the hull's finite physical budget and decides, in seconds and in one plain sentence, which payloads come aboard, where they go, who wins when two want the same thing, and what is shed when a generator trips.

> **Status (Sat 23:50 PDT): the engine (D1 to D5, the 200 ms safety table, estimator, tick loop, replay) and the screen run end to end with `uv run sdv demo`; 1712 tests pass.**

Run it

Python 3.12 via uv: `uv sync` once on Wi-Fi, then everything runs offline; `uv run pytest` runs the 1712 tests.

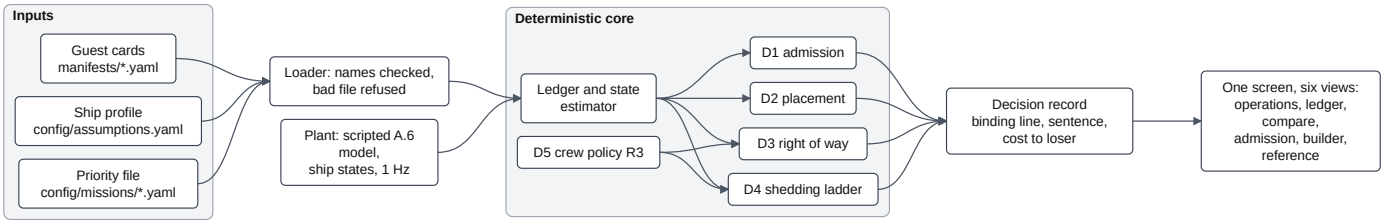
```
uv run sdv admit G4 --installed G1,G2,G3 --mission survey
uv run sdv admit G4 --installed G1,G2,G3 --set G4.hangar.mass_t=0.8
uv run sdv admit G4 --installed G1,G2,G3 --set G4.consumes.power_kw.peak=50
uv run sdv admit G7 --card path/to/G7.yaml --installed G4,G5 --mission disaster_relief

uv run sdv run --mission sar --log-dir runs/b-sar && uv run sdv replay runs/b-sar --mission sar
uv run sdv demo
```

`sdv admit` runs D1 and D2 on the real engine: verdict, binding line, every ledger line, placement, ranked yes-if options and the one-genset-down check, timed against the 30 s ceiling. `--set` is the jury's edit: any number on any card, re-validated; an edit that cannot be applied or fails the schema is refused with the field named and the previous card kept. `--card path.yaml` judges a card handed over as a file (schema or vocabulary problems refused by field; unregistered activities judged on quantities, positions and relations, and the output says so); a bad file in `manifests/` is reported once and skipped. `sdv run` plays Scenario B through the tick loop and prints each decision's sentence; `sdv replay` re-runs the logs without the simulator and checks every snapshot and record byte for byte. `sdv demo` is the one demo command: the engine and the screen on http://127.0.0.1:8000 (`--port` if busy), Scenario B live at x12 with six views (Operations, constraint ledger, Mission compare, Admission review with the jury's edits of masses and nominal or peak kW, the profile and rule builder, Reference), a SIM bar (hold, pace, jump to minute 8, 12 or 19, restart, swap the priority file) and a crew chip (14, 6, 0 aboard). Every run the screen shows is logged under `--log-dir` (default `runs/live`) and the terminal prints the `uv run sdv replay <dir> --mission <file>` line for it.

Architecture

One ledger, one manifest language. Everything that uses or supplies a resource is described the same way and counted in one ledger of margins across the brief's eight classes: power, heat rejection, deck positions, mast positions, mass and stability, data links, compute and response time. Every decision names the ledger line that bound it. Host commitments are never shed; guests are admitted, placed, throttled or shed; crew are hands and authority.



Everything in the diagram runs. Card conditions are typed: every signal, position and capability name is checked against the ship profile, and an unknown name refuses the file (`src/sdv/loader.py`). Faults

enter only as debounced, confirmed plant signals, a genset trip on 2 of 3 signals, a pump fault on the alarm plus flow or delta-T (`src/sdv/estimator.py`); while a feed behind an over-commitment is stale the ladder stays armed and the record says so. Safety commands (stop, hold, secure; 200 ms, A.8) are answered from a table the core rebuilds every tick (`src/sdv/safety.py`), never by a solver. A replay of the frame and input logs reproduces every snapshot and record byte for byte; wall-clock latency goes to a separate timing log (data contracts). Prior art: a ship's power management system already trips consumers in a fixed order on a generator loss; ArgOS makes that ladder mission-aware and explainable.

Decision classes

Components live in `src/sdv/`; D1 to D4 read one ledger (`ledger.py`), D1 to D5 write the same decision record (D2's inside D1's) and every sentence is built from its fields (`explain.py`); ceilings are A.8's.

Class (ceiling)	Component	What it optimises
D1 Admission (30 s)	`admission.py`	Yes, yes-if or no in the mission envelope (rule below); the binding line is the most-over line, else the tightest of mass, stability, power, heat, the two mission lines, the three data links and compute; for yes-if, the minimal removal sets ranked by mission value lost, mission-critical last
D2 Placement (30 s)	`placement.py`	Lexicographic, by enumeration of every allowed layout: no installed part left without a position, most parts placed, fewest port moves, preferred positions first, then a fixed order; no margin term, no rejected-alternatives record (CON-18)
D3 Right of way (2 s)	`arbitration.py`	Opens when G4's active hard limit (below 20 %: land within 12 min) meets G3's survey leg, on the two cards' typed ship-state conflicts; the priority file decides how G3 yields (pause and resume, or abort open-ended) and whether G4's charge is reserved; cost to G3 in survey minutes
D4 Degradation (5 s, 60 s)	`shedding.py`	By response class: 200 MS never shed, 5 S first, 60 S mission-critical one step a second; throttle before shed in each class, then `shed_order`, then least mission loss per kW; ladder precomputed every tick
D5 Crew as a setting	`core.py` (`_r3_policy`)	Per crew level (14, 6, 0), from `rules.r3`: confirm, standing plan, automatic or forbidden; a priority file may override a level, never to confirm at 0 (A.6); at 6 a recovery whose lead exceeds the minutes left is "not now" until the landing officer reports the zone clear; at 0 an unlatched deck forbids it (memo)
D6 Physical Account	`tools/physical_account.py`	Every assumed number with its reason and where the engine reads it, traced to file and line: docs/PHYSICAL_ACCOUNT.md

D1 rule. Judged at both gensets, the 200 kW reserve (A.1) and cooling at 30 °C (A.10); it must also survive one genset down by throttling to the 95 kW gate, otherwise yes-if, with the raw one-genset-down line always shown (DEC-35). In port, installed guests may move between bays, each move reported with its crane time (DEC-37). Data links are judged at the cards' minimum rates and compute at nominal (DEC-38), so a SATCOM or rack oversubscription has a line to name. The seawater tipping point is reported; D1 and D2 answer in 1 to 8 ms.

Deliverables

- **Prototype, one documented command:** `uv run sdv demo` (Run it), the screen on the live engine.
- **Manifest schema, five cards plus one of ours:** manifests/schema.json, G1-G6 (G6 is our desalination unit), how to add a card; validated by tests.
- **README (at most 3 pages), Physical Account (1 page), Scenario C memo (PDF, at most 2 pages):** this file, the account (from config/assumptions.yaml and the cards) and the memo; `tools/build_pdfs.py` builds each PDF and checks its limit.

- **Live demo, 90 s of Scenario B, then Scenario A with the jury's card:** run sheet. Scenario B runs G1, G3, G4 and G5, the organisers' load-out (DEC-63; G1 air-cooled on the mast, on the bus, not on the loop). Minute 8: D3 decides between the drone (G4) and the seabed vehicle (G3) under both priority files. Minute 12: a genset trips; the load (49.5 kW survey, 82.5 kW rescue) holds under the 95 kW power limit and D4 shows both armed ladders, no step needed. Minute 19: a labelled, injected pump fault makes cooling bind on the loop (81 against 75 kW, the bus at 82.5 of 95); D4 slows G4's charge to 10 kW (survey) or takes G5 to 8 kW (rescue) (DEC-40).

Repository map

Every path: docs/REPOSITORY_MAP.md; every requirement and how we address it: docs/REQUIREMENTS.md; decisions and who made them: WORKLOG.md.

Limitations (today)

- Admission answers for a ship alongside: yes, yes-if or no. Scenario A' (admission under way, the A.11 windows in each card's `installation.at_sea`, "not now and when", the 0-crew logical reconfiguration) is designed, not built. B.0 `part_of` (D2) and the cards' activity-conditioned `excludes` (D3) are enforced; `requires` is data D1 does not read.
- Over-consumption is metered, not detected or handled offender-first. D2 is lexicographic (no margin objective, no rejected alternatives, CON-18). D3 covers one contention, G4's recovery against G3's survey leg, only while G4's hard limit is active; the priority file decides how the loser yields.
- The -40 % loop at 30 °C is an information line on the Admission view, not in the verdict (DEC-36); both mission lines survive a genset trip by choice (A.9). Minute 12 holds by design: the load fits under the 95 kW ceiling with 0 steps; the record states the whole armed ladder and why the brief's envelope set does not arise (DEC-40, DEC-61); the executed shed is the minute-19 pump fault, labelled INJECTED.
- The plant is a scripted A.6 model (`tests/support/sim_stub.py`, labelled SCRIPTED PLANT · A.6 on screen), not a physics simulator; `SDV_SIM=module:Class` swaps another in; the team's own simulator is on `feat/ship-state-fsm`, not main. Scenario B runs at the brief's 24 °C seawater. Only the separate route-weather tool (`tools/deploy_map`) reads public data: cached Open-Meteo weather (CC BY 4.0) and Natural Earth coastlines (public domain).
- A jury card file is judged on the CLI (`sdv admit --card`), not uploaded on screen. The screen has no roles or access control. With nobody aboard, R3 trusts a deck latched clear at the last sweep. Numbers come from the brief's annexes and our labelled assumptions, never from a real vessel.

With another week

- Pier-side authoring: a card drafted from the equipment manuals with retrieval, every number cited into a provenance row (source, page, date), the loader as the gate that refuses an uncited figure, and a human signing the card before admission; never inside D1 to D6 or the sentence builder.
- Measured against datasheet: meter each guest's draw against its card, name the 10-20 % over-consumer, and schedule planned outages as capacity events the ledger sees ahead of time. Then Scenario A' with its windows, D2 with a margin term and rejected alternatives, D3 over any colliding pair, ArgOS measured on hardware near A.7's rack, more files and cards.

Declarations and licence

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